

TRAN SI NAM

Computer Vision | Machine Learning | Networks and Telecommunications Intern

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PROFESSIONAL SUMMARY

Senior Electronics and Telecommunications Engineering student with academic project experience in computer vision, deep learning, object detection, OCR, dataset quality review, image and video testing, model evaluation, LAN prototype testing, and reproducible technical documentation. Contributed to a team-built Vietnamese license-plate recognition pipeline and completed a YOLO11 graduation thesis. Seeking internships in computer vision, machine learning, computer networks, or telecommunications.

EDUCATION

Bachelor's Program, Electronics and Telecommunications Engineering 2022 to 2026

VNUHCM-University of Science | Faculty of Electronics and Telecommunications | Ho Chi Minh City, Vietnam

- Undergraduate thesis defended on August 15, 2026: *Deep Learning Application for Dangerous Person Detection in Public Areas*.
- Academic focus: computer vision and machine learning, supported by computer networks, data communications, and telecommunications engineering fundamentals.

TECHNICAL SKILLS

Programming and tools: Python, YAML, Git, GitHub, GitHub Actions, Git LFS, Typst, Tkinter

Machine learning and computer vision: PyTorch, Ultralytics YOLO, OpenCV, FastALPR, fast-plate-ocr, object detection, OCR, training, inference

Data and media: data collection, YOLO annotation review, FFmpeg, image and video preprocessing, augmentation, Roboflow, Kaggle, Google Colab

Evaluation: train-validation-test splits, precision, recall, F1 score, IoU, mAP50, mAP50-95, confusion matrices, error analysis

Networks and telecommunications: computer networks, data communications, LAN application testing, telecommunications engineering fundamentals

PROJECTS

Vietnamese Automatic License Plate Recognition with YOLO and OCR

Jan 2026 to Apr 2026

Seven-person course project | [Repository](#) | [Case study](#)

Python | Ultralytics YOLO | PyTorch | OpenCV | FastALPR | FFmpeg | Tkinter | Kaggle | Git LFS

- Contributed to a seven-person, end-to-end pipeline that combines YOLO plate localization, FastALPR and fast-plate-ocr recognition, text normalization, annotated image and video output, and a LAN whitelist demonstration.
- Reviewed train, validation, and test organization; identified anomalous images and YOLO-format labels; and cross-checked data.yaml paths for Kaggle reproducibility.
- Tested multi-format video and FFmpeg workflows, annotated outputs, codecs, and failure cases involving missing models or dependencies, low light, small or tilted plates, and OCR confusions. The team reported 99.45% precision, 99.37% recall, 99.45% mAP50, and 77.01% mAP50-95 on the validation set.

YOLO11 Dangerous-Person Detection in Public Areas

Apr 2026 to Aug 2026

Undergraduate Graduation Thesis | [GitHub repository](#)

Python | PyTorch | Ultralytics YOLO11 | Roboflow | Google Colab | NVIDIA Tesla T4

- Built a two-class object-detection prototype and curated 3,140 images with 10,595 labeled person instances, including 247 self-captured public-area images and hard-negative examples, using an 80/10/10 train-validation-test split.
- Trained and compared five YOLO11 variants with consistent 832-pixel inputs and held-out evaluation.
- Selected YOLO11s after it achieved 0.87 precision, 0.69 recall, 0.77 F1, 0.80 mAP50, and 0.64 mAP50-95 on 314 held-out test images.
- Packaged training and inference scripts, YAML experiment settings, tests, checksums, versioned artifacts, model and dataset cards, and reproducibility guidance; documented limitations and the requirement for human oversight.

ADDITIONAL INFORMATION

Languages: Vietnamese (native); English (technical reading and writing)

Availability: Internship-level roles in computer vision, machine learning, computer networks, network communications, and telecommunications